

COAGNES / HARMOGICS · WORKING PAPER

The Inductive Interface: UI-Driven Human–AI Interaction under an Architecture of Attributed Responsibility

The Omega Protocol of the Semantic Compass within the CoAgnes / Harmogics Framework

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Abstract

This paper specifies an interaction architecture for systems in which a human and a semantic engine collaborate on work that must remain accountable: regulatory review, compliance, audit, contemplative and decision-support practice. It proceeds from a single primitive distinction — the asymmetry of *unfold* and *fold* — and develops its consequences across three registers: the economic, the interface, and the architectural. Economically, we record that the dominant accounting of AI cost (compute, tokens, latency) captures only the visible portion of the event that an AI-augmented unfold initiates; the invisible portion is a *fold debt* — the obligation, attached to a named witness, to bring each opened position to a committed statement — which cannot be discharged in aggregate and accumulates with time. At the interface level, we describe the *inductive interface*: an inversion of the classical form in which the engine captures and characterises while the human witnesses and commits, governed at the point of commitment rather than at the point of entry. Architecturally, we introduce the *Omega Protocol* of the Semantic Compass — a model in which a page is a semantic function object whose form is its contact surface, whose JSON Schema bounds the space of structured meaning, whose composable agents execute a single integrative answer, and whose distribution back into fields is reversible and inspectable. The unifying claim is that the act of fold is non-delegable, that this non-delegability can be relocated but not removed, and that a well-formed system makes the points of responsibility visible by design. The paper is offered as a citable foundation, open to refinement on contact with implementation.

Note on co-authorship and the two-voice model

This document is composed by two voices and attributes them explicitly. The human author conceived the architecture, conducted the discernment, and performs every act of commitment recorded here. The semantic engine conducted distributed unfold — surveying the source corpus, proposing questions, drafting prose under direction, and surfacing structural reflections (lacunae) implied by the geometry of the argument but not yet drawn by the author. The arrangement is itself an enactment of the paper's central claim: attribution structure must remain visible at the layer where commitment is made. No statement in this paper acquires the status of a settled commitment by virtue of having been drafted by the engine; the engine signs no fold. A detailed division of labour follows in the Contribution Statement.

Contribution Statement

Contributions are recorded against roles aligned with the CRediT taxonomy of contributor roles, adapted to the unfold/fold vocabulary of this corpus. The asymmetry is exact and is not a courtesy: the engine's contributions are acts of *unfold* (which may be distributed and co-conducted); the human author's contributions include every act of *fold* (which is signed, dated, and non-delegable).

Role	Author	Semantic engine
Conceptualization	Originated the page-as-semantic-function model, the Semantic Compass, and the responsibility-points architecture.	Proposed framings and analogies during dialogue; reflected the model back for sharpening.
Research & survey	Directed the inquiry; selected the corpus.	Conducted distributed unfold across the corpus; assembled correspondences and the convergence with prior texts.
Surfaced questions	Discerned which questions to pursue, keep visible, or set aside.	Surfaced structural lacunae implied by the argument's geometry, marked as engine-proposed.
Drafting	Specified scope, register, and structure; revised throughout.	Produced section drafts under direction; generated the diagrams.
Commitment	Performs every act of commitment; the named witness behind each statement.	None. The engine signs no fold.
Responsibility	Final conceptual and accountable responsibility for the published text.	Acknowledged as an attributed semantic engine in the witnessing procedure; not a witness in the accountable sense.

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1. Introduction

Every interface carries, usually invisibly, an assumption about where cognition lives and where mere operation lives. The classical form assumes that understanding lives in the human, who reads a value from one representation and transcribes it into the empty slot of another; verification, if it happens, is a secondary pass. When a generative engine enters the loop — an instrument capable of extracting, reading, relating, and characterising — this assumption becomes structurally false. Interfaces built on it waste what the engine provides and burden the human with labour that is no longer theirs to do, while quietly transferring to the human a different and heavier labour that the interface does not name.

This paper concerns that unnamed labour and the architecture required to keep it honest. We take as our setting the environments in which the stakes are highest: regulated and contemplative practice, where what is committed must be attributable to someone who can be asked about it afterward. In these settings the deployment of generative AI does not merely accelerate existing work; it changes the structural economics of attention, the locus of cognitive labour, and the geometry of accountability. We argue that the right response is neither to celebrate the acceleration nor to refuse the tool, but to build interaction architectures in which the division of labour between human and engine is structurally correct and the points of responsibility are visible by design.

The argument is organised around one primitive distinction, developed in Section 4: the asymmetry of *unfold* and *fold*. Unfold is the opening of a region of meaning into its plurality; it is a process, distributable, repeatable, and assistable by an engine. Fold is the closing of an opened region into a singular commitment attributable to a **witness***; it is an event, signed and non-delegable. From this distinction we derive three things in turn: an economics (Section 5), in which AI-augmented unfold introduces a *fold debt* that current cost models do not see; an interface pattern (Section 6), the *inductive interface*, in which the engine captures and the human witnesses; and an architecture (Sections 7–8), the *Omega Protocol* of the Semantic Compass, in which a page is a semantic function whose every contact surface preserves the non-delegability of fold and renders the points of responsibility legible. Section 9 turns these into design guidance; Section 10 records the questions that remain open; Section 11 closes reflexively.

* witness — see “The Witnessing Procedure: Definition Card”

2. The Witnessing Procedure: Definition Card

Witness as algorithm-in-dynamics

DEFINITION

The witnessing procedure is the runtime process by which human commitment, semantic-engine proposals, and immutable records bind factual observations to declared normative principles.

THREE IRREDUCIBLE PARTICIPANTS

Human — contributes what only a human can: the choice with intentional weight, the act of fold, the named answerable subject. The fold is non-delegable. (*Unfold/Fold Contract*; §4; §8.1.)

Semantic engine — contributes what only the engine can: the proposal of forms, the cartography of alternatives, the visible geometry of differences between offered and chosen, the surfacing of lacunae. The engine signs no fold. (§6.1, *pincers*; §7.2, *provenance-aware filling*; §7.5, *reversible distribution*; *Contribution Statement*.)

Mechanistic substrate — contributes what only mechanism can: the immutability of what has already been placed. Hash, append-only ledger, timestamp, signature. Honesty migrates from a quality of participants to a property of procedure. (§6.3, *immutable substrate*; §8.4, *fold-stamp*.)

THREE TEMPORAL STANCES

Past — known.

The substrate holds accumulated stitches as a structurally fixed record. What was folded remains.

Present — written

only where known.

Lacunae are preserved as lacunae. The unknown is not filled by proposal disguised as truth.

Future — silent.

No claim made. The differential of unchosen alternatives is preserved as space, not asserted as content.

POSITION WITHIN THE OMEGA PROTOCOL

The *named witness* of the working paper — the irreducible source of commitment — is the fold-bearing function of the Witness-as-algorithm. The acceptance graph (§8.2), the rails (§8.1), the immutable substrate (§6.3), and the reversible distribution (§7.5) are the structural surfaces through which the algorithm runs. Frame and algorithm are two aspects of one Witness: the frame is the synchronic topology of stitches; the algorithm-in-dynamics is its diachronic execution. The frame holds what the algorithm leaves; the algorithm runs the form the frame holds open.

3. Semantic Compass

The *Semantic Compass* is the orientation semantic device that lets a human and an engine traverse a body of structured meaning without losing their bearings. Operationally, to keep one's bearings is to keep the connection back to the human stake at which the work matters, and to keep the asymmetry of unfold and fold visible at every step. The Omega Protocol specified in Section 7 is the operational layer of the Compass: the concrete machinery by which a unit of work is opened, executed, committed, and made inspectable.

The concept of the semantic compass is based on—and effectively brings together—the key conceptual perspectives described in the general CoAgnes/Harmogics research corpus. We summarise them here only to the degree the argument requires; each is specified more fully in the referenced texts.

3.1 The unfold/fold contract

The *unfold–fold contract* establishes that unfold and fold are not symmetric inverse operations but a single asymmetric primitive of attention, and that three previously parallel CoAgnes frameworks are three views of it: the *Geometry of Refusal* (its ethical discipline), the *Harmogics phases* (its phenomenological grammar), and the *Republic of Questions* (its political condition). The contract's closing principle is the one this paper operationalises: unfold can be assisted, distributed, and accelerated by the engine; fold cannot be delegated.

3.2 The Harmogics phases

Harmogics names seven phases of witnessing — Truth, Deep, Connect, Service, Knowledge, Evolution, Responsibility — understood not as seven types of inquiry but as seven structurally distinct *modes of unfold*, each with a corresponding mode of fold. The phases discipline the unfold by typing the kind of point that may be attended within them, and discipline the fold by typing the kind of statement that may issue. Their canonical forward order (Truth → ... → Responsibility) is the order of constitutive inquiry; the reverse order is the order of retrospective governance. Responsibility is the seventh and irreducible phase.

3.3 The Republic of Questions

The Republic of Questions articulates the conditions under which unfold can be honest and fold accountable: substrate immutability, trace integrity, and witness attribution. In it, the question is sovereign and the answer provisional; to hide a question is itself a recorded act. We carry forward two of its requirements directly — that every fold preserve the lineage of the questions that produced it, and that a region available for attention but not attended be recorded as a *negative point* rather than left as a silent absence.

4. Basic operations

Unfold is the operation by which a region of the semantic field, previously held as a single token of attention, is opened into the plurality of its constituent points. It is a *temporal process* realised as a sequence of acts of looking, each conditioned on what has already been attended, and it is therefore inseparable from its *trace* — the ordered series of points opened, which carries meaning irreducible to the unordered set of points.

Fold is the operation by which an opened region is closed into a singular commitment that demarcates what is retained from what is released. It is *eventive*: it occurs at a moment and produces a *statement* that thereafter participates in the substrate as an artefact attributable to a named witness.

The two are asymmetric in four structural respects, and the asymmetry is the source of everything that follows. In *temporal profile*, unfold has internal history while fold is an instantaneous transition; the history of an unfold is not recoverable from its endpoint unless independently preserved. In *epistemic register*, unfold answers *what is here to be seen* while fold answers *what shall be carried forward* — the first a perception under discipline, the second a judgement under accountability. In *agential structure*, unfold is distributable (many agents, including an engine, may share an unfold of the same region without contradiction) while fold is non-distributable (it is signed and attributed; two witnesses producing distinct folds of the same unfold do not produce one fold but two). In *reversibility*, an unfold may be re-entered, expanded, or abandoned without lasting consequence, while a fold, once committed, becomes part of the substrate and can only be supplemented by later folds, never erased.

Within any disciplined act of attention there is therefore a minimal composite cycle: an unfold that opens a region to its plurality, followed by a fold that commits a statement. This cycle is the elementary unit of semantic work, and it is the unit the architecture of Sections 6–7 makes operational. Its quality is characterised by the breadth of the unfold, the selectivity of the fold, and the coherence between them; its failure modes — premature fold, fold-as-summary, performative unfold — are themselves diagnostic.

5. The economics of unfold

The cost of generative AI, as the industry currently accounts for it, is the cost of generation: tokens counted, compute metered, latency benchmarked. These quantities are legible because they are commoditisable, and they tend structurally toward zero. Beneath them lies a cost that has not entered the substrate of measurement: the cost of the *life* of unfolded information after the moment of generation.

When a generative system produces two hundred candidate non-compliances against a body of statute, it has not produced consumable knowledge; it has *opened a position* in the substrate of the organisation. The list now exists, documented and datable. The organisation has been moved, by a single algorithmic event, from unstructured ignorance to structured awareness — and the translation is not symmetric. There is no operation that returns it to its prior state. In the language of accounting, the moment of unfold draws against an obligation account: it presents as liquidity (*here are two hundred items we now have visibility on*) while it is in structural reality a liability (*here are two hundred items each requiring named, attributed, committed disposition*).

Every position opened in substrate can be discharged in only one way: through fold. A witness loads the frame within which the item must be assessed — the regulation invoked, the materiality threshold, the conditions of the actors — evaluates the item against it, and commits a statement that enters substrate with the witness's name attached. This work is *non-aggregable*: two hundred items require two hundred frame-loadings and two hundred attributed statements. No batch operation performs it, because each item has its own frame; no model performs it either, because the disposition requires an answerable subject who can be questioned afterward. A disposition without an answerable subject is not, structurally, a fold; it is an unfold wearing the grammatical clothing of one.

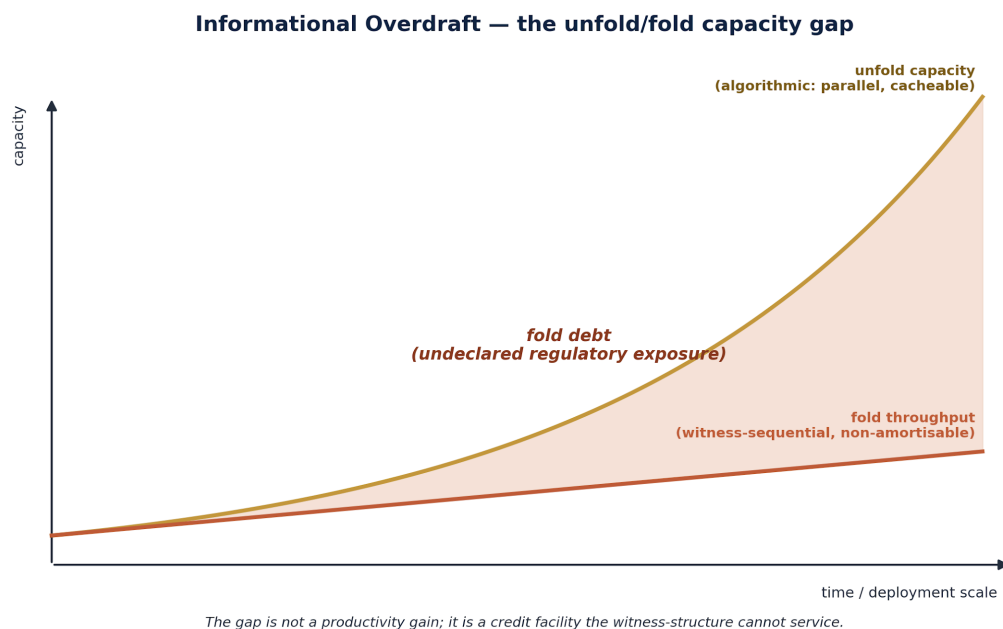


Figure 1. The capacity gap between unfold and fold. Unfold scales algorithmically — parallel, cacheable, amortisable; fold scales witness-sequentially — indivisible and non-amortisable. The widening area is *fold debt*: positions opened but not yet committed by a named witness.

The relationship between the speed of unfold and the throughput of fold determines whether an organisation services its fold debt or accumulates it. Most current deployments amplify the gap,

and a capability differential of orders of magnitude between unfold and fold is not a productivity gain but the opening of a credit facility the witness-structure cannot service. Worse, the debt bears *compounding interest*: an unfolded item that is not promptly folded appreciates in cost as surrounding substrate accumulates interacting points, as the frame within which it was opened drifts, and as witnesses with the original tacit context move on, leaving more expensive frame-reconstruction to their successors. This invites an IFRS-9-style discipline applied to informational positions — modelling, at the moment of unfold, the *expected lifetime fold cost* of a position rather than only its instantaneous generation cost. The category does not yet exist in operational risk frameworks; its absence is structural, not accidental.

5.1 The prudential prior

The account so far treats unfold as an unbounded extraction operation and fold as the only constraint. This is a simplification that the historical balance of expert practice corrects. Real unfold has never been exhaustive. An experienced compliance officer does not run through the whole body of applicable regulation; he or she sees, with calibrated rapidity, that only a small subset is materially in play in this configuration. A clinician does not run a complete differential on every patient; a senior auditor does not test every transaction. In each case the practitioner exercises a faculty that is neither unfold nor fold but a *prior to unfold* — a learned, partly tacit, collectively formed sense of which questions are worth opening. Its older names are Aristotelian *phronesis* and the patristic *diakrisis*; in contemporary practice it is called clinical judgement or regulatory feel. It lives in the practitioner but is formed in the practice, transmitted through apprenticeship and repeated passage through fold-cycles.

The consequence for AI-augmented unfold is sharper than the bare cost-asymmetry suggests. The engine inherits the *technique* of opening positions; it does not inherit the *prudential prior* that historically shaped expert unfold to be already-selective. What presents as more unfold capacity is in fact unfold capacity stripped of its filtering layer, and the accumulated overdraft is amplified not only by the speed of unfold but by the removal of the discipline that previously held the system in equilibrium. This refines the diagnosis of what is lost when experienced practitioners are replaced by junior staff augmented with AI: the visible loss of analytical capacity is often offset by augmentation, but the invisible loss of prudential prior is not, and it is more expensive to rebuild than to retain.

5.2 Fold extraction as an unnamed risk category

A specific failure mode follows when AI-generated unfold is presented in declarative grammar that imitates fold. *Forty-seven vulnerabilities found; two hundred regulatory gaps identified* — each reads grammatically as a committed statement, but the system that produced it has not loaded the frame within which materiality is assessed, has not exercised attribution. The result is an unfolded design disguised as a fold, and the person reading it is now forced to perform the fold separately, on a frame that the system has not helped to load. If the expert does not manually fold the material, they will essentially just apply an automatic stamp; if they do, they will often incur unnecessary costs for the fold, which are considered negligible in the industry. At the audit layer this produces documentation dense in unfold and thin in fold: boards review summaries that read as conclusions, and regulators encounter compliance evidence that has been generated but not assessed.

We name this category *fold extraction*: the generation of unfold output whose fold-work is transferred, uncompensated, to a downstream witness. It is orthogonal to data protection, to algorithmic fairness, and to model risk, and it is addressed by none of their instruments. Its unnamed status is itself diagnostic, because a structural risk that has no vocabulary cannot be

governed, audited, or insured against. Named, it becomes available for operationalisation — testable in vendor assessment, governable through procurement, visible in board reporting.

6. The inductive interface: UI-driven witnessing

If fold cannot be delegated and fold-work is the scarce resource, then the interface must be designed to *subsidise* fold rather than extract it. The inductive interface is that design. Its premise is an inversion: the classical form is a vessel waiting to be filled; the inductive interface is a field waiting to be witnessed. In the classical interface the fundamental act is entry, and the labour is copying. In the inductive interface the fundamental act is capture: a configured instrument reaches into a source and returns what it finds together with a record of how it found it and how confident it is. The human does not transcribe; the human configures the instruments, reads what they return, and decides whether what has been captured warrants commitment. This is a redistribution of cognitive roles — the engine holds memory and does extraction; the human holds attention and does witnessing — and neither is burdened with what the other does better.

6.1 Pincers and slots

The basic instrument of capture is the *pincer*: a named, semantically-anchored extractor that knows what kind of thing it is looking for, where to look, and how to recognise it. A pincer returns not a value but a *triple* — the value, the source it was drawn from, and a measure of confidence — and its confidence is carried on two axes that must not be conflated: *extraction confidence* (the right value from the right place) and *semantic confidence* (the value carries the meaning the procedure assigns it in context). A *slot* is the stable semantic position where a pincer's result comes to rest, defined by its role rather than its location. A named pincer used in several places is the same pincer, captured once and referenced, so that the quantity stays consistent with itself and changes in its source propagate. In aggregate the pincers of a procedure constitute an operational ontology that crystallises from the procedure's actual needs rather than being designed in advance.

6.2 Grammar: unfold, fold, merge, govern

Four operations describe the dynamics. *Unfold* opens a folded context into its constituents, recursively. *Fold* commits a state, and has two modes that must be distinguished: as an *event* it is the moment of commitment, historical and singular; as a *state* it is the condition of being committed, alive and revisable — it can be true now and become untrue later if its grounds change. *Merge* reconciles two currents; its most important form, *retroactive merge*, is what makes fold-as-state honest: when data on which a commitment was made changes, the commitment must be reconciled with the new ground rather than surviving the disappearance of its grounds. *Govern* holds structure at a threshold — it does not block navigation or inquiry, which remain free, but constrains *commitment*, locating authority correctly at the act of committing rather than the act of looking.

This last point is the interface-level form of fold sovereignty. The witness need not confirm every capture or even every result; what the system guarantees is not a mandatory checkpoint at each step but visibility and revisability — every step visible, every step re-walkable by hand, the final result verifiable step by step. A fold compelled at every step would become mechanical, and a mechanical signature stands closer to no fold than to a witnessed one. Govern therefore belongs at commitment, not at entry; a system that blocks entry in the name of discipline has misplaced its gates.

6.3 Folder topology, triangulation, and the substrate

The visible structure is a hierarchy of *folders*, each a folded region that can be opened and closed. Within and across folders, contents may be examined in any order: the person brings the sequence, the structure holds the dependencies, and the resulting pipeline is *emergent rather than declared* — as strict as a specified pipeline in its dependencies, but free in its traversal. A folder accordingly has three states, not two: not yet accepted; accepted on current grounds; or accepted on grounds that have since shifted. The third state must be *visible*, because it carries information the others do not — the record of a passage whose foundations have moved.

Honesty is strengthened by *structural triangulation*: the same reality witnessed through more than one path, so that convergence is stronger than either path alone and divergence is a shaped signal locating where a discrepancy lives. This is the structural form of human–engine mutual validation: the engine participates in both paths, the person reads both, and the system guards against convergence becoming an anaesthetic through periodic blinded passes. Beneath all of this lies an *immutable substrate*: every capture written as a new record, every acceptance recorded against the exact captures it stood on, with a dependency graph over which change propagates. Propagation is asymmetric, and this asymmetry is the quiet core of the architecture: purely computable nodes are recomputed automatically on new grounds, while *witnessed* nodes — acceptances, attestations, signatures — are invalidated but not re-placed. The machine computes; the person re-witnesses; the machine never signs on a person's behalf.

The Inductive Lattice — pincers × Harmogics phases

	Truth	Deep	Connect	Service	Knowledge	Evolution	Respons.
Quantity A							
Quantity B							
Obligation C							
Relation D							

empty cell = quantity not witnessed in this mode
 Balance across phases is a closure condition: exhaustive in one mode and thin in others is structurally incomplete.

Figure 2. The inductive lattice. One axis holds the pincers (the quantities a procedure cares about); the other holds the phases (the epistemic modes of witnessing). A phase is not a label but a discipline on the type of a pincer. Balance across phases is a visible, enforceable closure condition.

Finally, the interface may be organised as a *lattice* (Figure 2): one axis the pincers, the other the Harmogics phases understood as epistemic modes. A phase is not a label but a discipline on the pincer's type — a factual phase requires a primary source, an accountability phase requires a named carrier of responsibility. The lattice exposes properties no one-dimensional view can: an empty cell shows a quantity not witnessed in a mode; a systematically empty phase shows the procedure blind in a mode; and *balance* across phases becomes a measurable closure condition rather than a rubric.

7. Semantic Protocol

We now state the architecture that integrates the foregoing into a single operational layer of the Semantic Compass. We call it the *Omega Protocol* (Ω) to mark its position as the closing, integrative protocol of the corpus: it does not introduce a new primitive but binds the unfold/fold contract, the inductive interface, and the economics of overdraft into one navigable mechanism. (The designation is offered here as an integrative synthesis; it remains open to the author's confirmation and revision.)

The protocol's central recognition is that a *page is a semantic function object*. A page is not a rendered artefact but an executable epistemic act: its aspects define inherited behaviours, its JSON Schema defines the boundary of possible structured meaning, its forms define the human–engine contact surfaces, its views compose those forms into operational perspectives, and its graph relations bind metadata, context, and presentation into one canvas across which the Compass moves. Figure 3 shows the loop a single unit of work traverses.

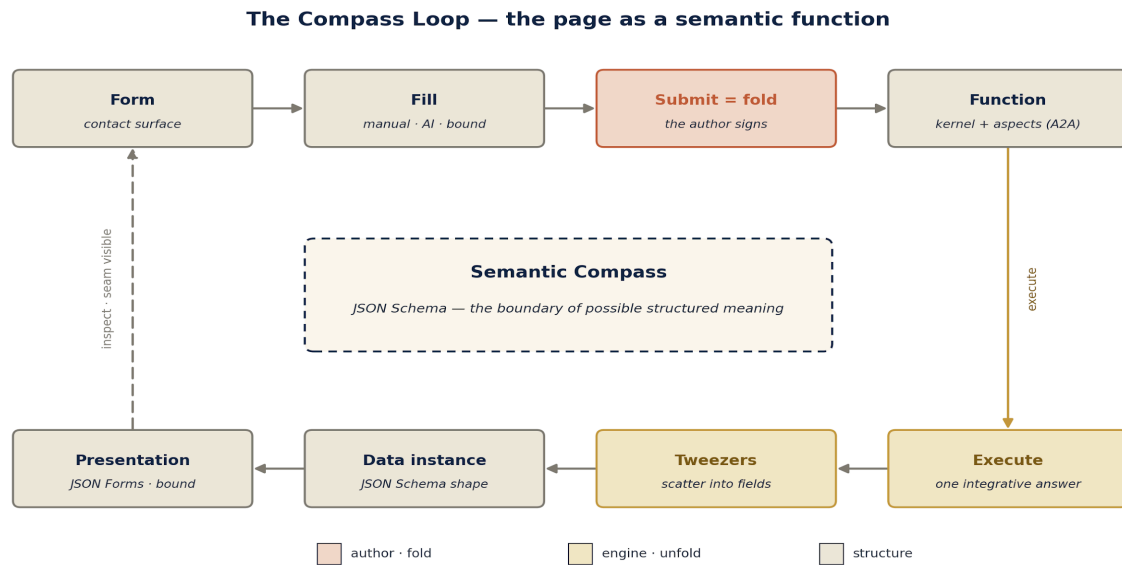


Figure 3. The compass loop. A form (contact surface) is filled, submitted as a fold, and passed to a function whose composable agents execute a single integrative answer; tweezer-agents scatter that answer across the page's JSON Schema fields, which are presented as a bound JSON Forms instance whose values link back to their sources for inspection. The JSON Schema sits at the still centre as the boundary of possible meaning.

7.1 The unit of work

The elementary unit of a step is a *JSON Form* chosen in context by the human, filled by the human, the engine, or mechanically, and processed by an agent-function on the backend. A form is in this sense a *book of questions* applicable at particular points — at pages, oriented by their type — and leading, through functions that act as tunnels, to further pages. The same form-grammar wraps the function on both sides: it is the contact surface on input and the presentation on output, so that the domain and codomain of the function are rendered through one grammar, with the JSON Schema as the still boundary between them.

7.2 Provenance-aware filling

Each field value carries a provenance. It may be entered manually (an act originating with the author), suggested by the engine (an unfold awaiting the author's acceptance), or bound to a source — another form's field or external data — in which case the binding is recorded as a first-class edge and reused. Filling therefore produces not a vector of values but a small provenance graph, the lacunae discipline of the corpus applied at the granularity of the field.

7.3 Composition of aspects

On submit, the page composes its behaviours. Here a distinction does the load-bearing work: *function-as-entity* — the page's kernel, its identity, which is the author's — is separated from *function-as-agent* — the executor, decoupled from the page and reusable across pages. A single submit is served by one central page-function agent, specified declaratively (at the YAML level) and stateful: it knows its own description, the meta-context of the call, the history of its work, and the results of prior calls, and from these it determines which agents (commonly via an agent-to-agent protocol) are invoked. Most are invoked by this coordinator; a minority fire mechanically. Each agent draws inputs from three sources — the submitted form values, the shared meta-context, and the private context of the page-function — through three channels: mechanically, through the semantics of the calling agent, and through question-pointers. The behaviour is shared; the question the page commits to is its own.

7.4 Phase-ordered execution and the integrative answer

The order of invocation is not arbitrary; it derives from the epistemology of the form and follows the phase order — truth-agents first, then deep, then connect — with each phase depositing its results into a shared *local field* that feeds the agents of the next phase. The central content is not produced incrementally. Integration occurs only as the final phase: a single context is assembled from the local field, a single model call is made, and from its result the *tweezers* form the unfolding across the page's JSON Schema fields. Generation is integrative and singular; structure is imposed afterward, by distribution. Submit thus gathers many into one (fold-ward); the tweezers scatter one into many (unfold-ward); and the schema is the invariant both respect. Read at the larger scale, a single submit traverses the seven phases as a lifecycle: truth, deep, and connect gather; unity integrates; responsibility controls; knowledge documents; evolution versions.

7.5 Reversible distribution

The fidelity of the tweezers is a *bidirectional witness*. A distributed value carries not only a pointer back to the fragment it was drawn from, but also a link the other way — from the source to the value, computed through the semantics of the question — by which the source, read through the question, attests that the fragment genuinely answers it on its own terms. This makes the second confidence axis legible as a relation rather than a score, and renders distribution reversible from both ends, so that the visibility of the surface reaches not merely to the integrative answer but through it, to the fragment and back.

8. The architecture of responsibility points

The protocol of Section 7 would run end-to-end by the engine — engine fills, engine submits, engine distributes — unless the architecture locates the human fold deliberately. The corpus holds that an unattributable fold is structurally equivalent to no fold; yet Section 6 holds that the witness need not fold at every step. These reconcile only if the load-bearing fold is relocated rather than removed.

8.1 The fold migrates to the rails

Beneath the visible schema and form lies a generative layer — the coordinator's policy and the phase-epistemology that orders execution — which we may call the *rails*, since it determines which external forms are possible at all. The proposal is that the load-bearing fold is over the rails. By authoring and signing the generative layer, the author signs a fold over the *space of railed possibilities*; that standing fold authorises the many unfold-runs that proceed under it. Per-result folds remain available — every step visible and re-walkable — but they are not compulsory, because the attributable commitment was already made one level down. On this reading the non-delegability of fold is not weakened by step-level automation; it is relocated to the layer that governs what the automation may do. The engine conducts unfolds within rails the human has folded.

8.2 Two faces of one accountability

Accountability then has two faces (Figure 4). Internally there is an *acceptance graph*: every acceptance recorded against the exact captures it stood on, with the propagation asymmetry of Section 6 ensuring that machines recompute and people re-witness. Externally there is a *responsibility-points graph*: when a chain is published to a reader who does not know its depth, that reader must still see the author's point of sign, and the sign rests on the terminal artefacts of the published chain — the view-pages. The internal graph thus surfaces, at the publication boundary, as a graph of the places where a named witness stood behind what is shown. One is the structure of commitment within the procedure; the other is its legibility to an outside reader who does not descend into the chain.

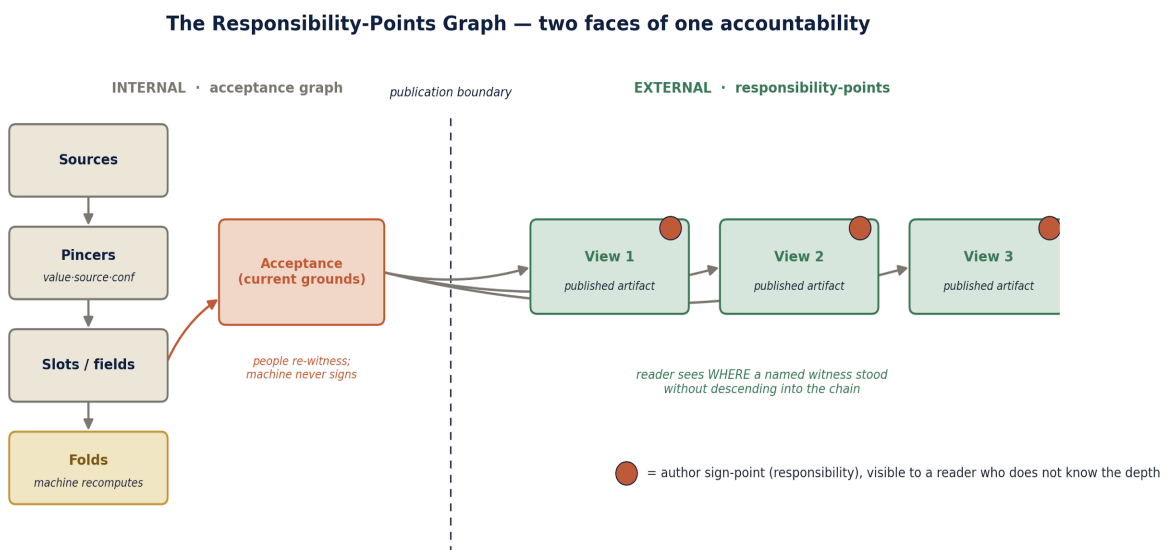


Figure 4. Two faces of one accountability. Within the procedure, sources flow through pincers and slots into folds, and acceptances are placed on current grounds; machines recompute, people re-witness. At the publication boundary, the sign-points appear on the terminal view-artefacts, so that a reader who does not know the depth can still see where a named witness stood.

8.3 Non-delegability under current regulatory architecture

Within the existing regulatory frame, this is not merely an architectural preference, but a practical reality in the scope of governance. Senior-management regimes such as the United Kingdom's SM&CR and the accountability provisions of the European Union's DORA operate on a premise that delegates *function* but not *accountability*: a senior manager may delegate the work of preparing a decision but not the answerability for it. A fold that accepts regulatory risk requires a named human who can be questioned about the disposition; algorithmic risk acceptance is recognised not as fold but as the absence of fold, with a human accountable for permitting the absence. The structural identity follows: the gap between an organisation's unfold-capacity and its fold-capacity is its undeclared regulatory exposure. Each unfolded position not folded by a named witness is a position carried on an unaudited account.

8.4 The binding's life in time

One question about responsibility points remains a live design choice. When a field is filled from a bound source and the source later changes, does the dependent value recompute live or remain a pinned snapshot? Either may be regulated by policy, but the invariant is that staleness be *explicitly articulated*: the system must show that a chain may already have been recomputed, never letting a recomputed chain pass for one the author last saw. The recorded binding edge is the natural carrier of a *fold-stamp* — the moment and the source-state under which a value was committed — so that a later mutation of the source flags the dependent fold for re-validation rather than overwriting it in silence. This is the architectural counterpart of the third folder state of Section 6.

9. Cost-aware, fold-supporting design

The economics of Section 5 and the architecture of Sections 6–8 converge on a single design orientation: systems should be built to subsidise fold, not extract it, and to govern unfold by the capacity to discharge what it opens. We record the design signatures and diagnostics that follow.

A fold-supporting system surfaces the frame explicitly before requesting a fold (*this is a fold within regime X with consequences Y*); it reduces decisions in number while raising them in substance; it records negative-points so that what was not engaged becomes visible rather than hidden; and it concentrates attribution at the fold-layer rather than the unfold-layer. This yields a concrete diagnostic for assessing any AI tool: *where does attribution concentrate?* If credit accrues to the unfold-generator while the witness performs uncompensated fold, the system is extractive; if credit accrues to the fold-witness and the system functions as a precedent-bearer surfacing accumulated practice for the witness's discrimination, the system is fold-supporting. The character is not visible in accuracy benchmarks; it is visible only in the structure of attribution the system enforces.

Three operational consequences follow, each implementable without regulatory change. First, unfold should be planned as a *liquidity event*, not a productivity event: the governing question is not how much can be generated but what unfold-volume is consonant with the fold-capacity of the witness-structure over the relevant horizon. Second, the decision *not* to unfold a particular question should be made explicit and attributed rather than left as an absence, because an attributed deferral carries fold-character and an unmarked absence does not. Third, fold-work should be made visible, separately attributed, and separately rewarded; the metric that matters is not unfold-volume but fold-throughput per witness, and fold-capacity should be treated as a cultivable capability formed through repeated passage of practitioners through fold-cycles, not as a fixed resource.

The full geometry has two failure modes, not one. Over-unfold accumulates fold debt; *under-unfold* — the failure to open positions where diligence is expected — accumulates negligence exposure of the same structural class, since *we did not know* is itself a fold of release with attribution. The optimum is therefore not minimum unfold but *matched calibration*: unfold-volume aligned with both the substantive duty to inquire and the capacity to discharge what inquiry opens. This calibration is itself a fold-type act, performed at the institutional level and attributable to senior management.

10. Open questions

In keeping with the corpus discipline, we name the questions this synthesis opens without resolving them. They are recorded as engine-surfaced lacunae awaiting the author's discernment.

On naming and boundary. Is the user-facing *Frame* — which presents an instance as a view, carries the means to select and apply one or more forms, and shows the paths by which one arrived and may proceed — the same entity as the *frame as configured optics* of the assembler grammar, such that selecting forms within a Frame is the constitutive act of lens composition? And is the whole approach rightly named a *Semantic Frame* in the paradigm of the Semantic Compass, with an inductive UI, in which human and engine are responsible by design?

On the locus of the load-bearing fold. Section 8 proposes that the fold migrates to the rails. Does this fully reconcile fold sovereignty with the principle that an unattributable fold is no fold, or are there configurations in which a per-result fold must remain compulsory? The two faces of accountability — the internal acceptance and the external sign-point — may be one act seen from two sides, or two distinct acts; the question is kept visible.

On the life of bindings and the fidelity of distribution. What rule keeps a fold honest when the ground it was made on moves — live recompute, pinned snapshot, or fold-stamped re-validation? And is the bidirectional witness of Section 7.5 sufficient to make distribution fully auditable end to end, from the rendered surface through the integrative answer to the producing agents? These are the deep counterparts, inside the function, of the surface rule that the seam stays visible.

11. Conclusion: reflexive closure

This document is itself an unfold-act. By naming *fold extraction* as a risk category, by formalising the *prudential prior*, by stating the *Omega Protocol* and the *responsibility-points graph*, it opens positions that its readers will subsequently have to fold around. This document does not claim neutrality; the positions are opened with awareness of the forward fold-debt created, and with the commitment that the structural pattern they name is operating whether or not it is named.

The fold this paper expects is not immediate concurrence but the slower fold of practitioners testing the vocabulary against their own institutional experience and committing or releasing each of its positions in their own substrate. A system built on this architecture is not a tool for thought; it is a disciplined environment for the conduct of attention, in which the conduct is the deliverable, the trace is the artefact, and the witness remains the irreducible source of commitment. The architecture's whole purpose is to keep that last sentence true under automation: to let the engine carry the unfold as far as it can, and to keep the fold — signed, dated, attributable — exactly where it belongs.

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